Paul Parisot

Software Engineer - System Programming

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About

Systems-focused software engineer with strong foundations in C/C++, Luau, Go, and low level languages.
Experienced in building programming languages, game engines, and distributed systems.
Open-source contributor with 10k+ LoC on critical packages, and deep hands-on experience with the Roblox platform: Luau scripting, EditableMesh procedural generation, and a from-scratch C++ engine reimplementing Roblox client/server replication.
Currently deepening theoretical CS foundations (graph theory, complexity theory, automata) through preparation for the French *agrégation d'informatique*, a highly selective national CS competitive exam.

Skills

Technical: C/C++, Golang, Python, Flutter, JavaScript/TypeScript, Haskell, Lua(u), OCaml, Assembly
Systems: Runtime Library Loading, Network Replication & Serialization, ECS Architecture, Reflection (C++26)
Other: Git, Devops (Docker, Kubernetes), CI/CD, Linux

Professional Experience

Gno.land - Junior Software Engineer (Open-Sourced)

- Developed 6+ core packages in Gnolang (Go variant), including printf, EVM compatibility, and a petition system.
- Found and fixed **6 critical bugs** causing runtime crashes; **10k lines of code** contributed.
- Wrote extensive tests under strict CI/CD and code review policies.

Faurecia / Forvia - Software Demo Engineer

- Built a **Unity-based demo app** showcased at **CES 2024**, simulating real-time GPS and UI navigation.
- Designed interactive tools like a **GPS path engine** and POI search interface.
- Delivered production-ready features **ahead of schedule**, praised for speed and code quality.

Epitech / IONIS - Technical Assistant & Instructor

- Taught **400+ students** in C, C++, asm, and Haskell; reviewed and graded assignments.
- Created **interactive learning content** (quizzes, flipped classes, tutorials).
- Applied **pedagogical methods** (Constructivism, Cognitivism...) to boost student engagement.
- Contributed to a historically high **success rate** in ASTEK-coached cohorts.

Personal Projects

Luau / Roblox Platform Engineering

Built a **PNG decoder & renderer from scratch in Luau**, parsing raw PNG binary data (chunks, zlib/DEFLATE inflation) and rasterizing pixels directly on-platform with no native image APIs.
Designed a **procedural generation engine** (islands, houses, trees) using Roblox's low-level **EditableMesh** API to generate and optimize triangle geometry at runtime.
Deep working knowledge of Luau's sandboxing model, scripting constraints, and Roblox Studio's mesh/rendering pipeline.

Skills: Lua(u), Procedural Generation, Binary Parsing, Computational Geometry, EditableMesh, Roblox Studio

blorox – Modular 3D game engine inspired by Roblox (link)

Designed a game engine in C++ that reimplements Roblox's client/server architecture, with rendering and logic modules **dynamically loaded at runtime** (including a runtime-loaded `lua.so` for embedded scripting).
Currently tackling **network replication**: serializing partial mutations of C++ class fields (single-field diffs) between client and server without full-object re-serialization.
Prototyping a solution using **C++26 reflection** to auto-generate per-field replication metadata; project paused pending compiler support maturity.

Skills: C, C++, Lua, ECS Architecture, Runtime Linking, Network Replication, Serialization, Engine Design

cclass – Runtime Memory Mutation Engine (link)

Parses the process's own compiled binary at runtime and uses `nm` symbol lookups to compute exact function sizes.
Copies raw function bytes into an executable memory page (via `mmap`) allocated directly after each object instance.
Reserves the last 8 bytes before the copied function's end as a hidden slot for the object pointer, read by the function itself as an implicit `this` – emulating method dispatch without a vtable.

Skills: C, Linux Systems, x86 Assembly, mmap, Memory Safety, Low-level Debugging

Sledge – Cross-Cloud Data Migration, Synchronization & Orchestration Engine

Architected and shipped a generic cross-cloud migration backend in Go, via a custom cloud abstraction layer (`provicol`) supporting AWS S3 and GCP.
Utilized the Docker API at runtime to dynamically spin up, isolate, and orchestrate concurrent environment containers.
Applied strong relational algebra principles to optimize **SQL queries** and manage cross-platform data integrity.

Skills: Go, SQL, Docker API, Python, Data Integrity, Cloud Computing, System Automation

ZZZcript – Statically-typed FP language (link)

Built a **lexer, parser, AST interpreter** from scratch, with a recursive eval engine in **Go** inspired by Haskell.

Skills: Go, Haskell, Parsing Theory, AST Design, Functional Semantics, Interpreter Architecture

Education

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Master's level diploma)
Level-7 Certification - Expert in Software Engineering

McGill University, Montréal 2025 – 2026
Certificate in Management
Coursework Accounting, Business Administration, Organizational Management